



METHODOLOGICAL FRAMEWORK FOR FINDING HATEFUL CONTENT

**Schapelle Corby - Update
Presentation**





20-year sentence for tourist in drug case that gripped Australia

An Australian tourist was sentenced to 20 years in jail yesterday for attempting to smuggle more than 4kg (9lb) of marijuana into the Indonesian holiday island of Bali, concluding a trial that has gripped her country.

Schapelle Corby, 27, a trainee beautician from Brisbane, was also fined 100m rupiah (£5,800), although the end of the verdict was barely heard as the sweltering courtroom, crowded with family, friends, holidaymakers and an army of Australian journalists, descended into pandemonium.

Loud boos drowned out the cheers of a few anti-drug campaigners. Corby's mother, Rosleigh, stood up and shouted at the three judges: "Liar, liar. Honey, we are going to take you home."

Corby, who fainted once during the trial and has had to be treated for stress, cried briefly but remained relatively calm, although clearly stunned by the verdict. She mouthed the number "20" in Indonesian several times and then turned and told her mother: "Mum, it's OK, it's OK."

Her plight has galvanised Australia into an unprecedented outpouring of public sympathy for an alleged drug smuggler after defence lawyers claimed she was an unwitting victim of gangs who employ airport baggage handlers to stuff drugs into luggage.

It had been maintained that the drugs found in her unlocked surfboard bag were planted. The claim appeared to gain weight when it emerged that baggage handlers at Sydney airport had been involved in a drug smuggling ring that had been operating the day Corby flew to Indonesia.

Scores of people flocked to her defence after learning she faced the death penalty.

BACKGROUND

- Schapelle Corby case (2005, Bali drug conviction), a highly publicised and controversial case that generated polarised debate around justice, fairness, and media influence.
- Analysed online discussions by collecting publicly available user-generated content from platforms such as Reddit and X, where users share opinions and respond to media narratives.
- Large dataset allowed us to explore how discussions evolve over time, and how different perspectives are formed, shared, and contested within online communities.
- Provides insight into how public discourse develops beyond traditional media coverage.



WEB SCRAPING

**Scrapers extract publicly
available website
content.**



AIMS, OBJECTIVES AND EXPERIMENTAL METHODS

EXPERIMENTAL METHOD

- Platform selection
- Regulation review
- Data scraping (posts/comments)
- Data stored in JSON format

DATA PROCESSING & ANALYSIS

- Text cleaning & normalisation
- Tokenisation
- Scraping methods compared
- Evaluated using metrics

AIMS

Develop a framework to collect, label and store case-related data ethically

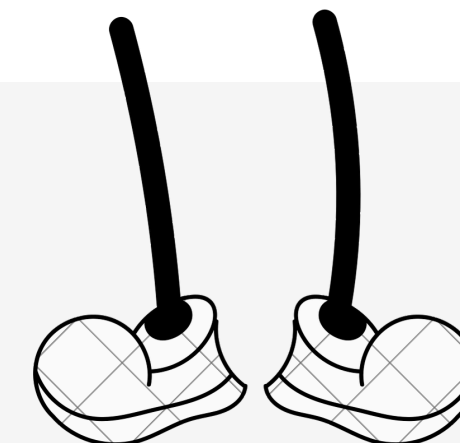
Evaluate tools under platform and ethical constraints



OBJECTIVES

Identify relevant online platforms (public forums)

Evaluate tools under ethical constraints



Project Progress Overview



Research & Planning

- Risk assessment
- Identifying hate speech terms on Schapelle Corby



Data Collection

- Testing scraping tools
- Tools tested:
 - No code: BrowseAI, Parsehub, Octoparse, Apify
 - Code: Python scraper



Data Cleaning & Preparation





HATE SPEECH CATEGORIES



Case-related

Hate Terms

Definition
Refers to offensive discourse targeting a group or an individual based on inherent characteristics and that may threaten social peace

Legality-based

Crime-based

Familial-based

Authority-based

Class-based

Moral condemnation

Intelligence-based

⋮
Argue guilt

⋮
Crime involvement

⋮
Targeting family

⋮
Targeting system

⋮
Social Judgement

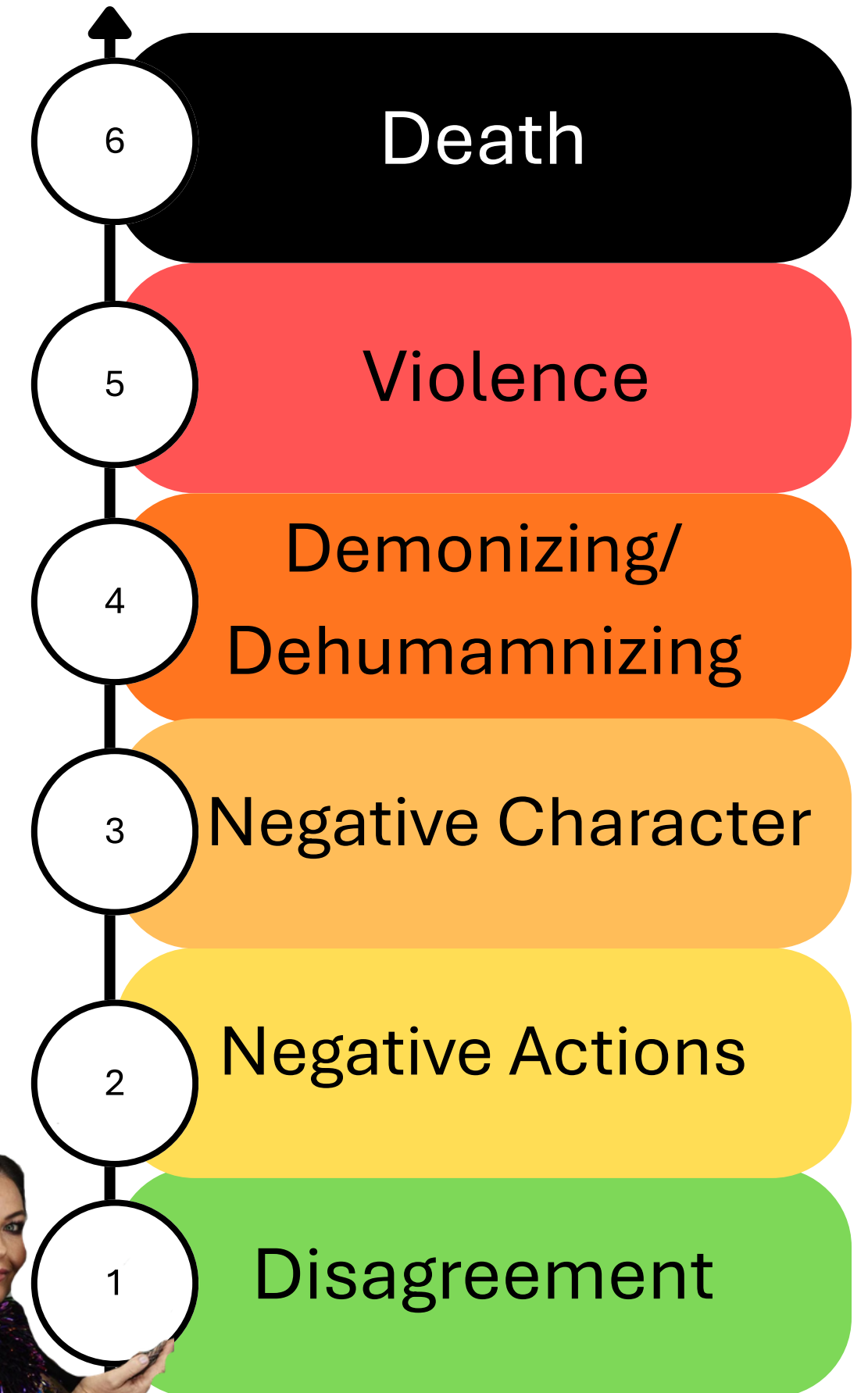
⋮
Judging actions

⋮
Direct Insult

MEASURING HATE SPEECH INTENSITY

Observations so far...

- Most comments with hate terms are likely to fall within Level 1 - 3 of the hate intensity scale
- A comment can contain multiple hate terms from different hate speech categories



PYTHON SCRAPER - TECHNICAL CONTEXT



Scraper

The scraper was built using Python 3.13.12 with integration of a Scrapy spider web-scraping framework and Trafilutra, a python text-extraction library that strips ads, menus and clutter

Early Challenges

When programming there were many issues when handling the scraping of different sites' HTML structure's. Google and Bing also don't allow scraping so DuckDuckGo was the search engine we used

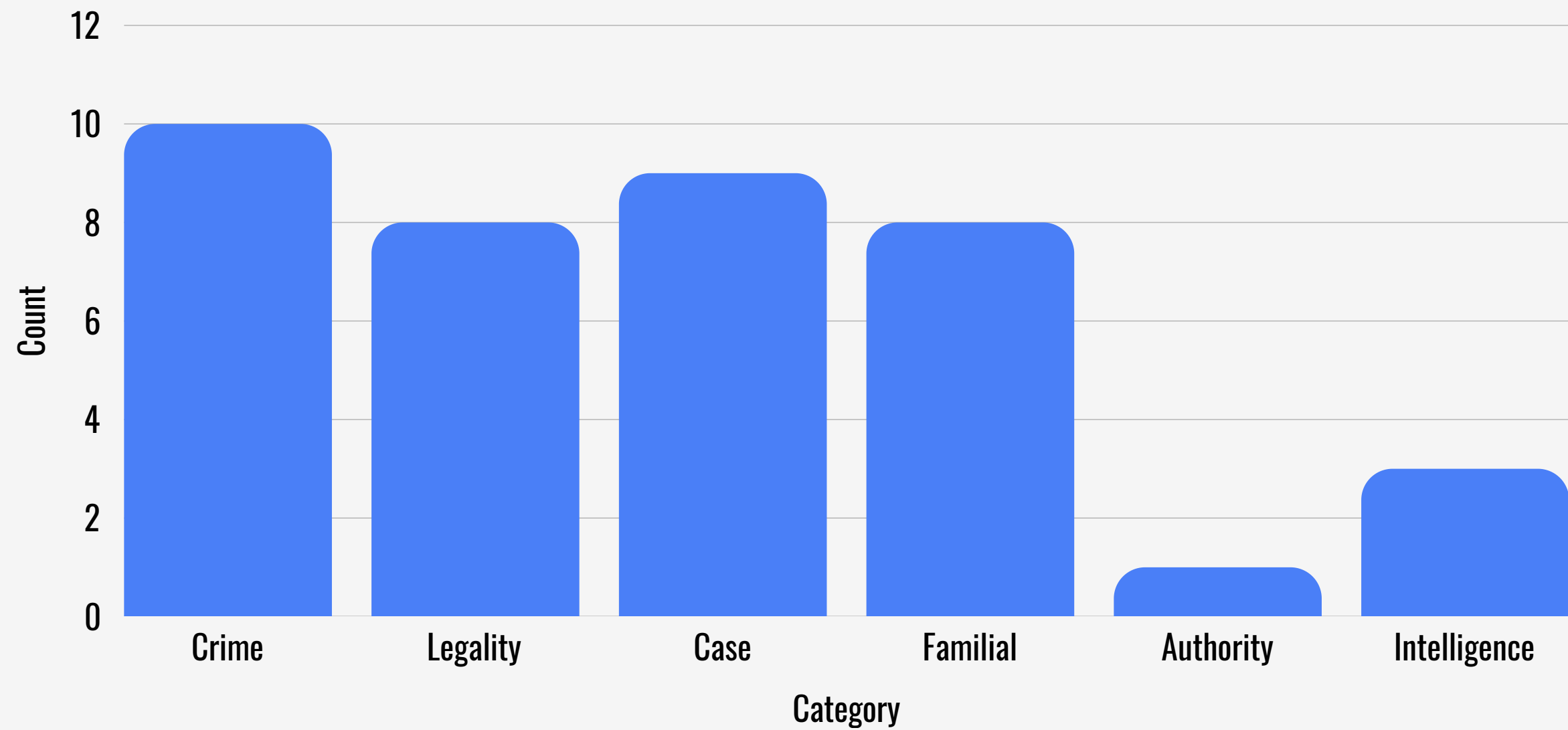


PYTHON SCRAPER - DATA COLLECTION

● ● Sites Scraped

- ◆ 9 News
- ◆ ABC
- ◆ Perth Now
- ◆ Win News

Bar Chart of Categories Collected from Python "Scrapy" Web Scraper



🔍 factual categories dominate due to news-ony scraping ✕



CURRENT LIMITATIONS

Future Work

- Reddit is the only social media with publicly available data collection
- Currently awaiting Reddit approval to begin data collection
- Once permission is provided: expand dataset, compare news vs public sentiment
- Stabilise pipeline for cross-platform scaling.



WEB SCRAPING METHODS COMPLETED

Tools Used:

- BrowseAi
- Parsehub
- Octoparse
- Apify

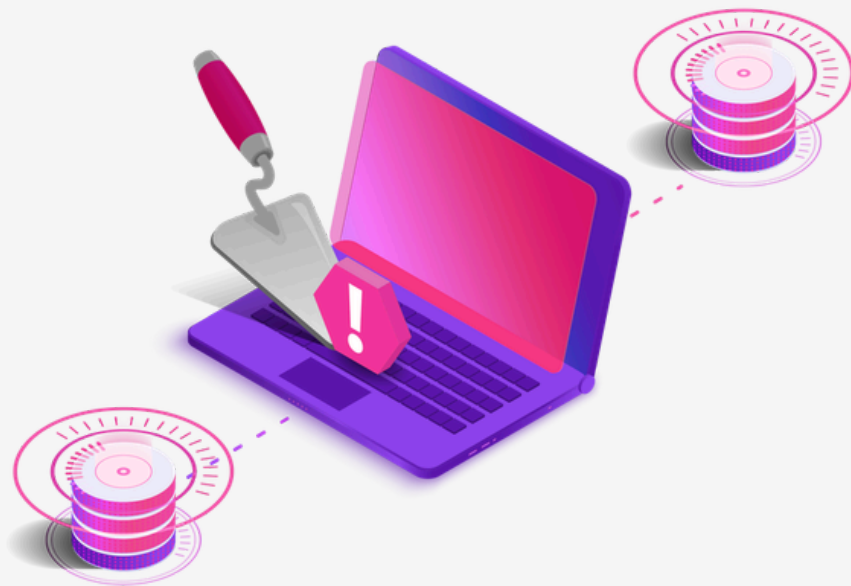


Process:

- Keyword-based searches related to Schapelle Corby
- Extracted publicly available comments
- Stored data in JSON format

Key Findings:

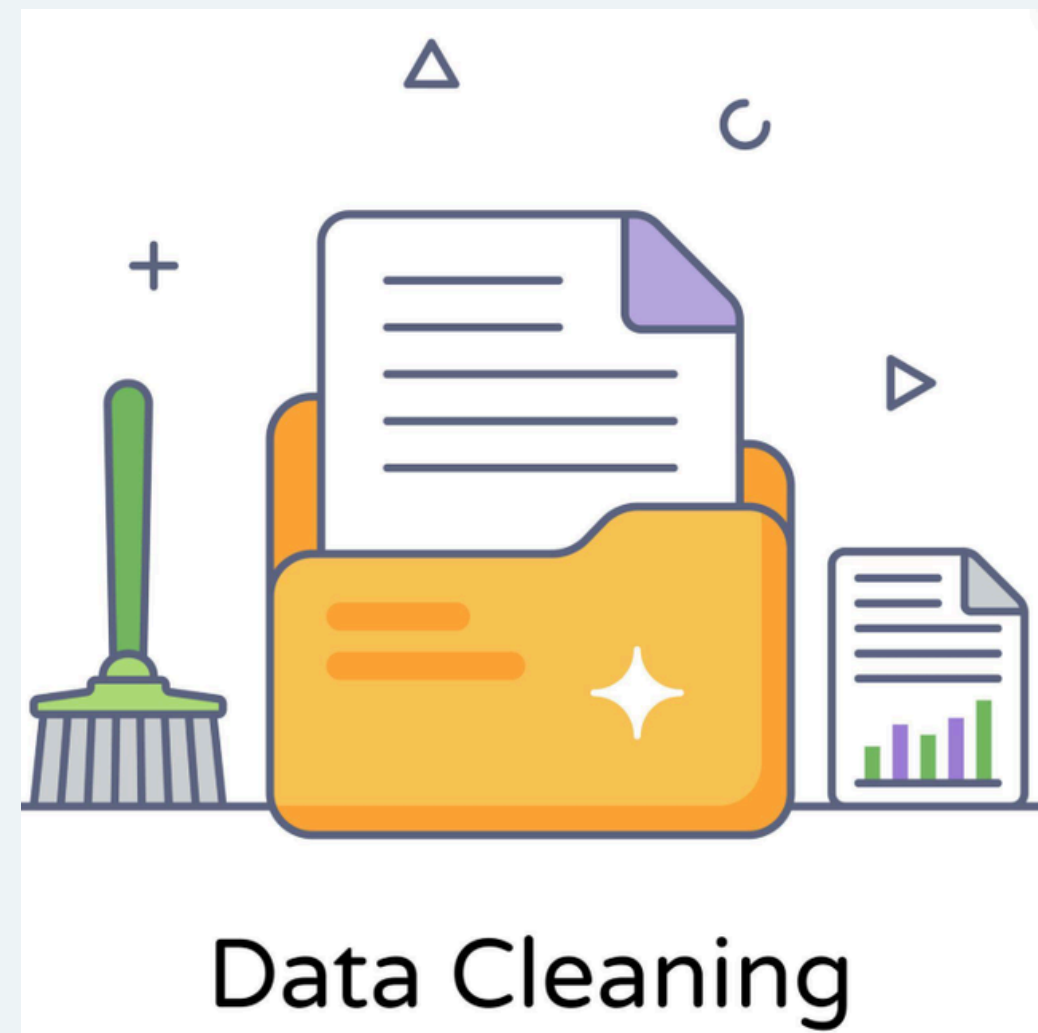
- Parsehub collected the most data
- BrowseAi was easy and simple to execute the tasks
- Octoparse and Apify collected the least amount of data



Data cleaning process

**Raw data collected
from multiple
scraping tools**

**Identified
inconsistencies and
noise in datasets**



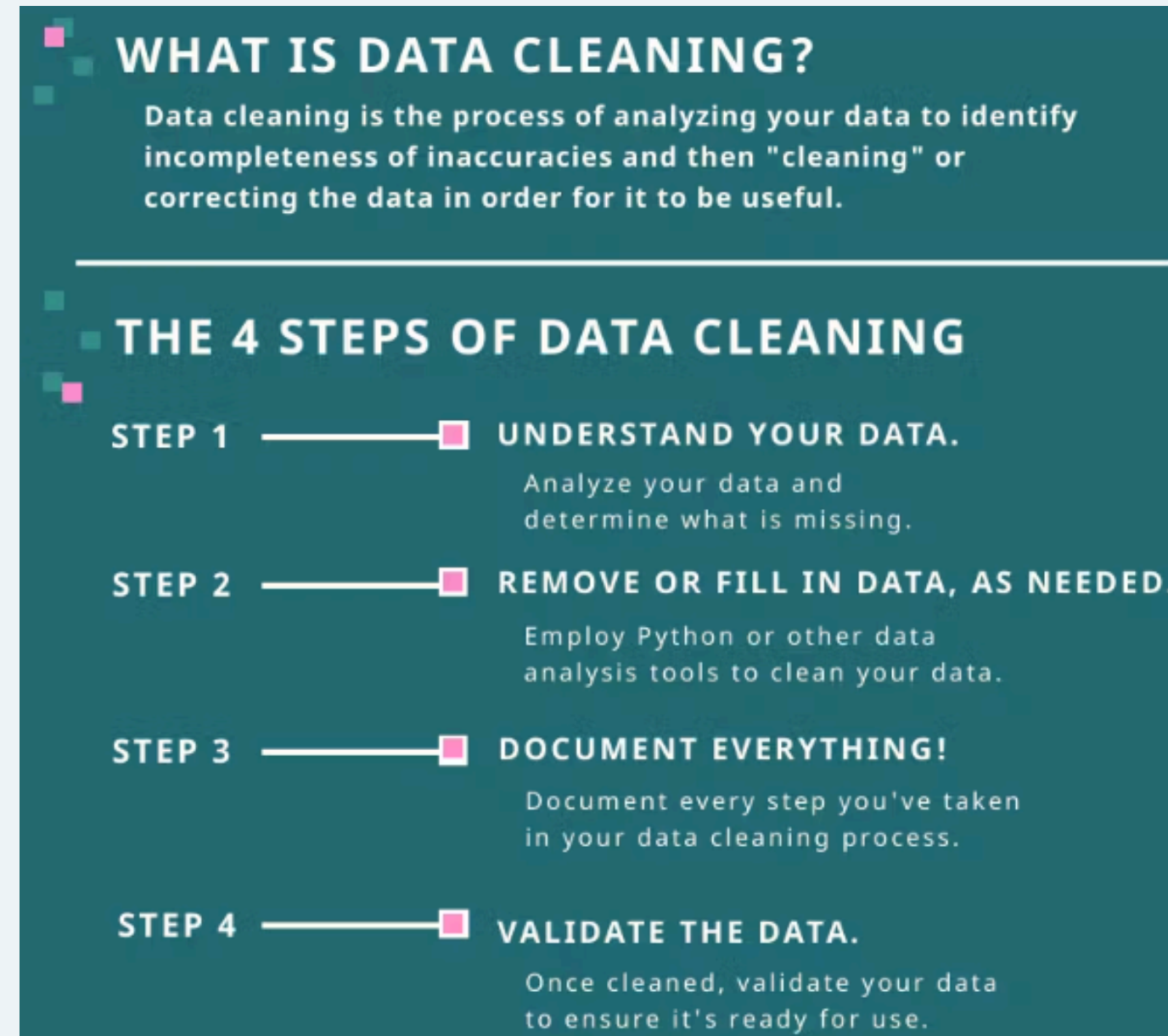
**Standardised and
cleaned textual data**

**Prepared structured
data for analysis**

Using Python for Data Cleaning

**Automated cleaning
using Python**

**Structured data with
Pandas**

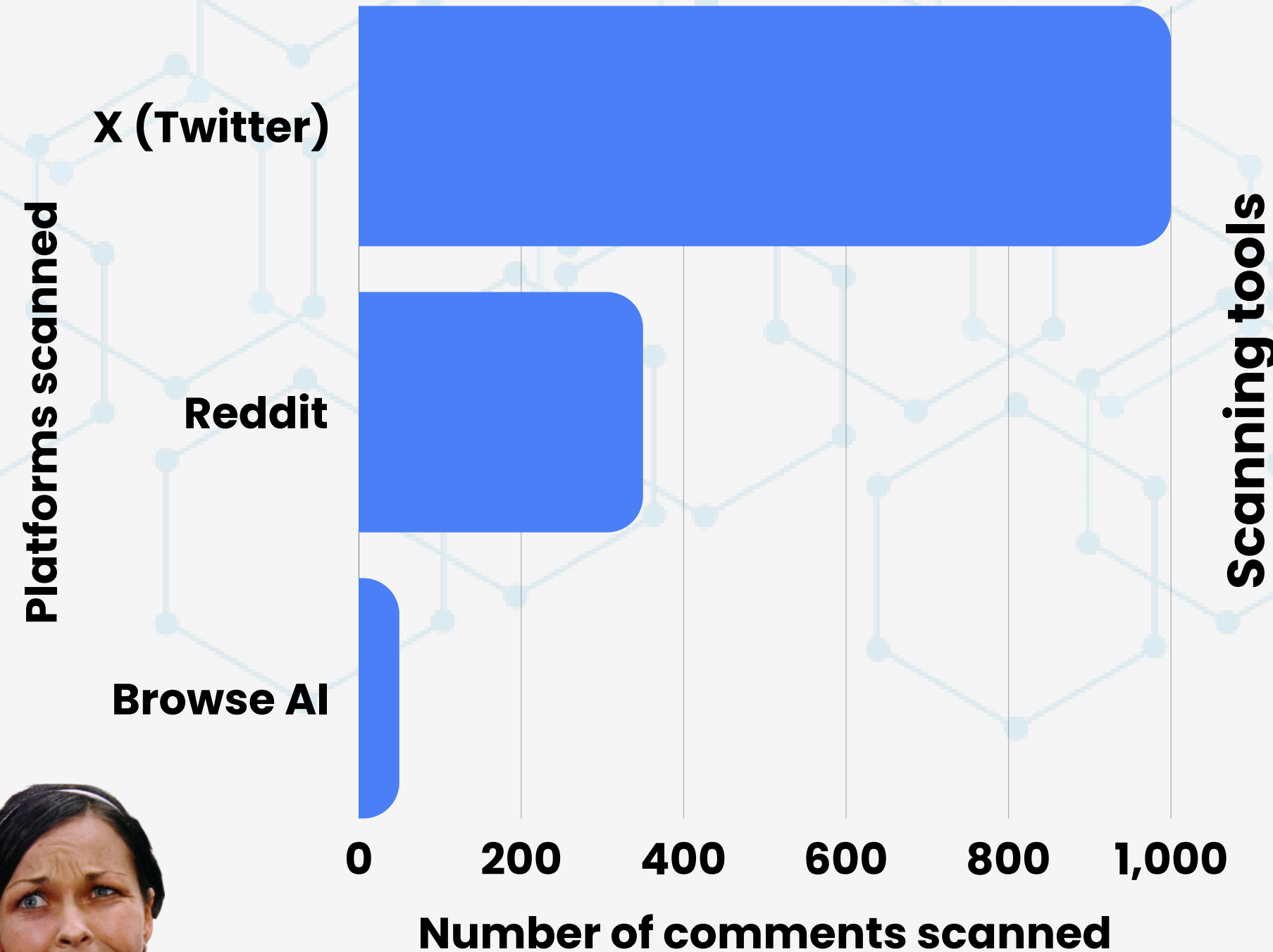


**Applied text
processing and
regex techniques**

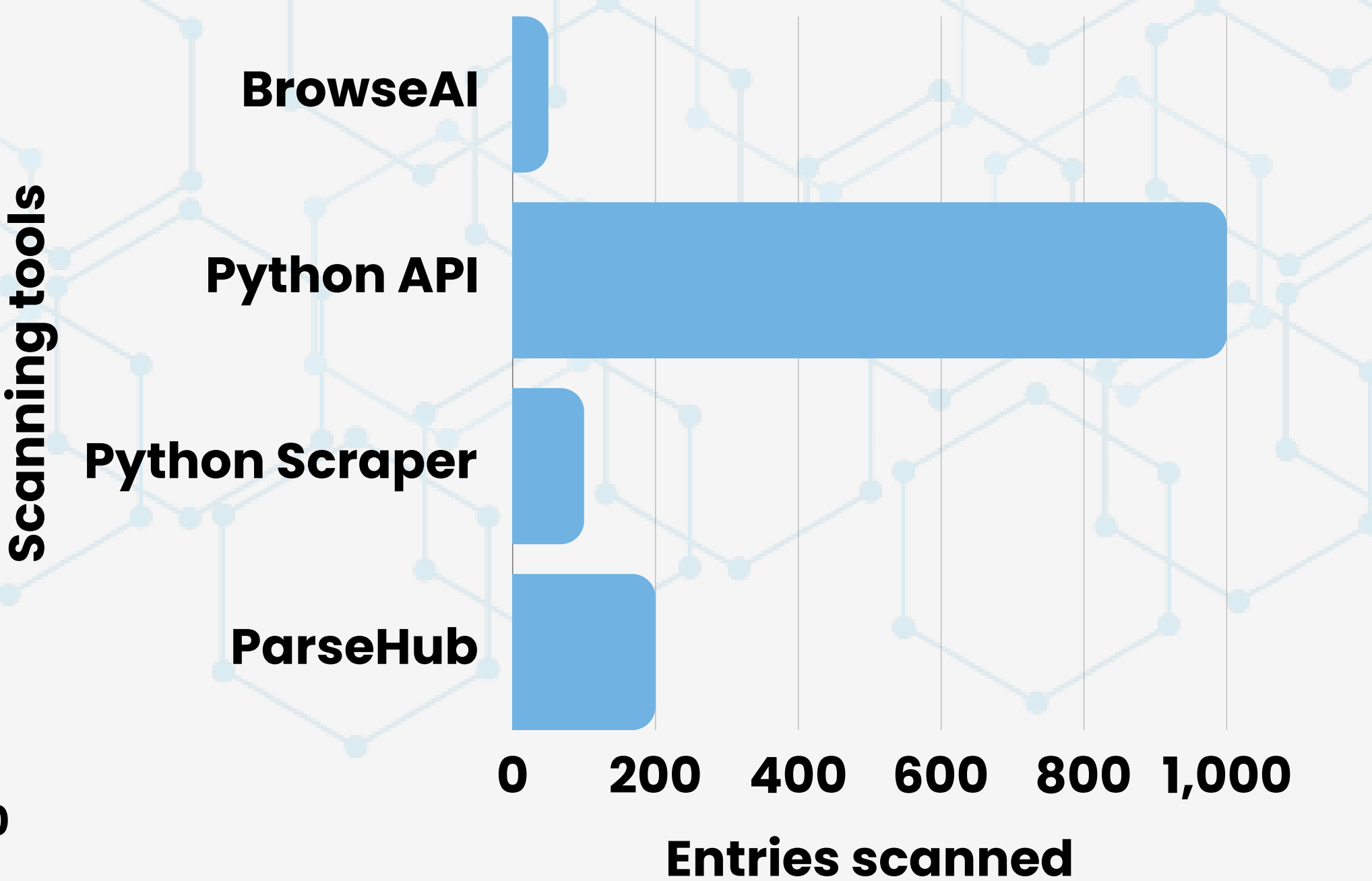
**Exported clean data
for analysis**

RESULTS AND ANALYSIS

Number of Scanned
Comments Across Sites

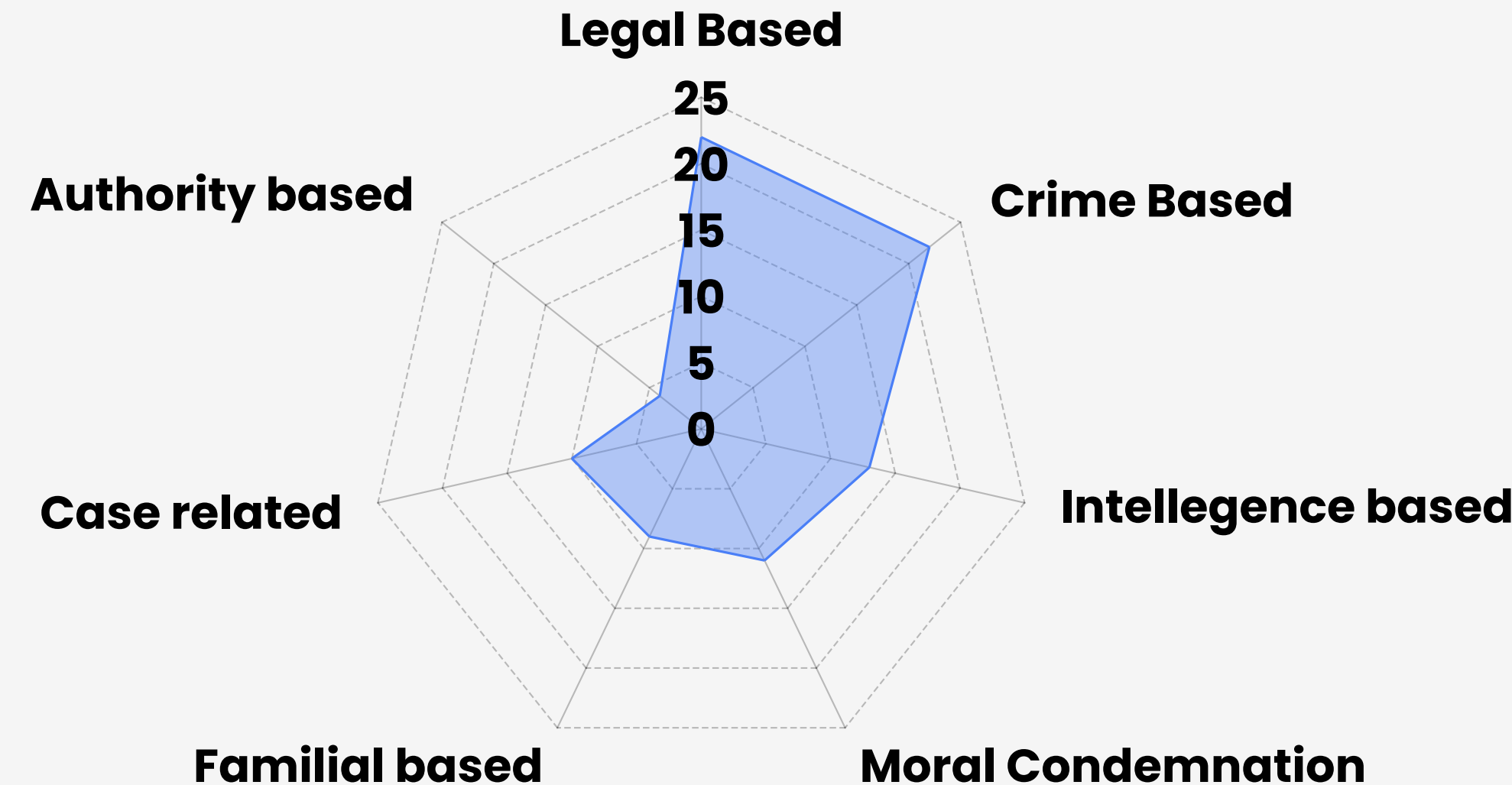


Total Entries of
Collected Data

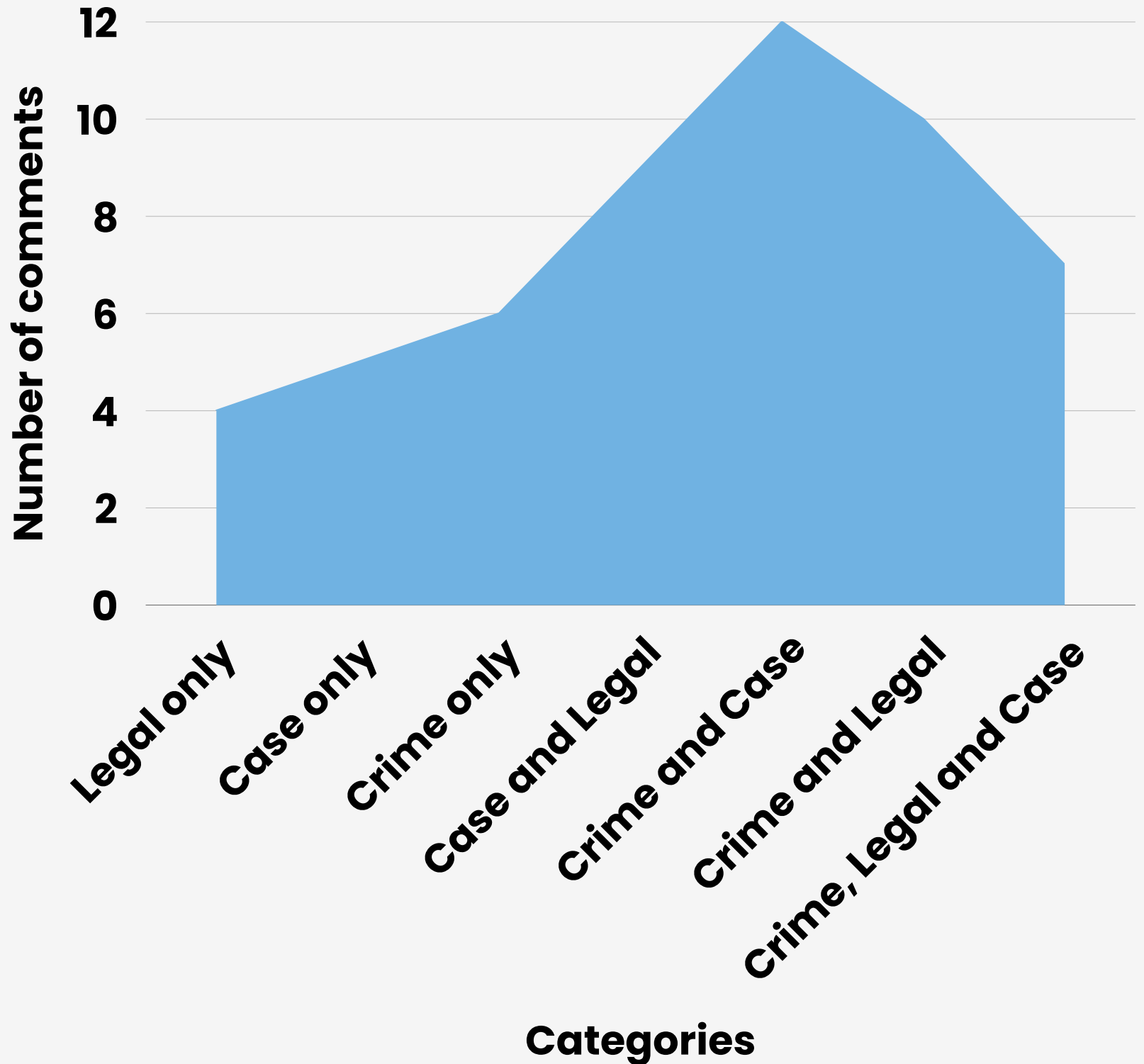


RESULTS AND ANALYSIS

Number of Comments Found in Each Category



Relationship Between the Number of Comments Found in Each Category

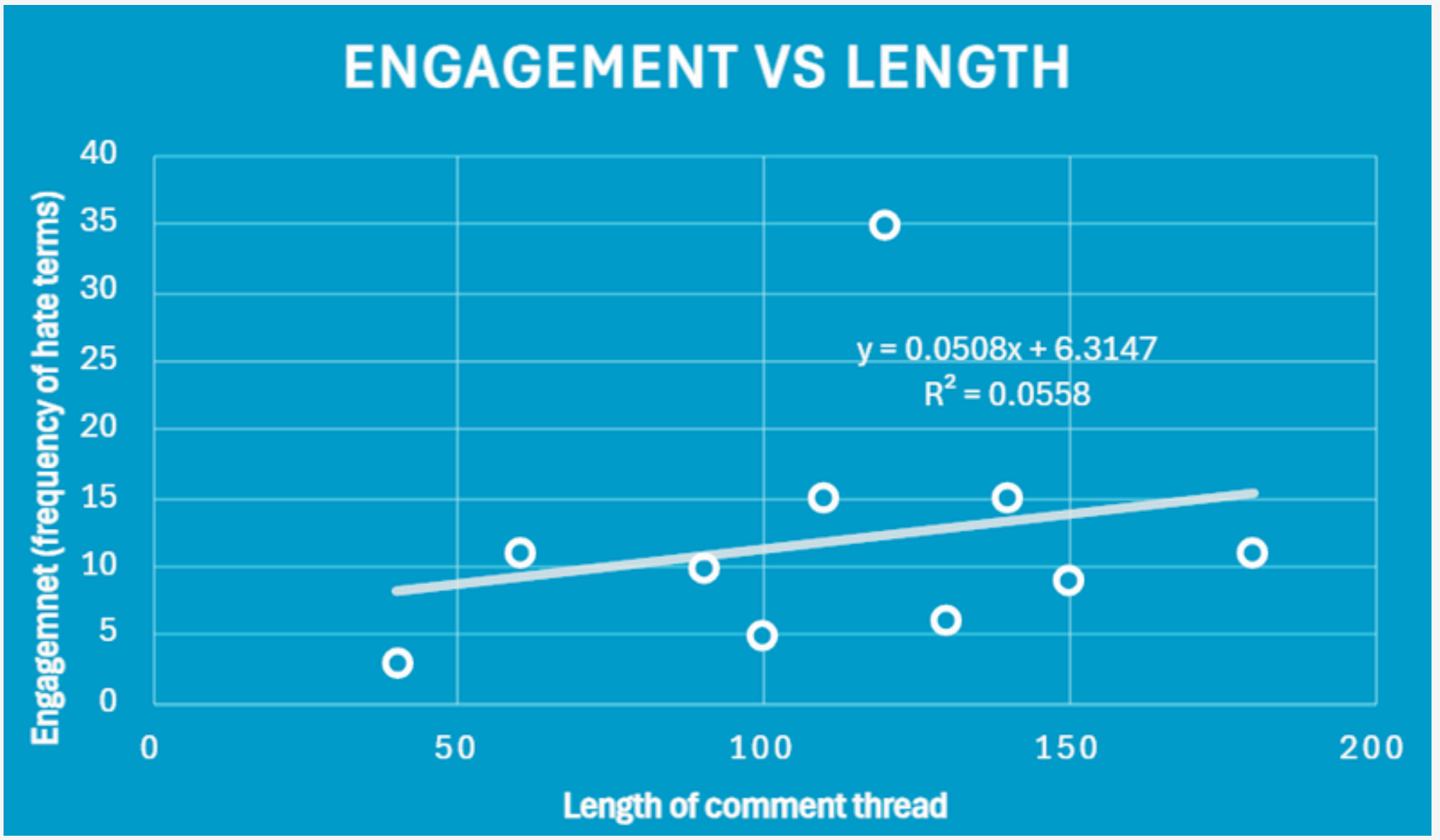


RESULTS AND ANALYSIS

Correlation of Types of Words Heatmap

frequency of words found together

	Crime	Legal	Case	Moral	Class
Crime	100	72	80	30	25
Legal	72	100	70	40	20
Case	80	70	100	35	15
Moral	30	40	35	100	10
Class	25	20	15	10	100



Key analysis:
The correlation of 0.0558 suggests a moderate positive relationship with the amount of comments and hate terms found

TIMELINE WEEK 8-12

WK 8



- Reddit API + permission
- Expand dataset (if approved)

WK 9



- Merge + clean data
- Standardise dataset

WK 10



- Apply categories
- Identify patterns

WK 11



- Develop hate scale
- Assign severity levels

WK 12



- Create graphs + analysis
- Finalise report

